Unit 1

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- Computer Network :
- A computer network is a system in which two or more computers and devices are connected together to share information, resources, and services.
- Purpose: To enable communication (email, chat, video calls), resource sharing (files, printers), data sharing (databases, cloud), and remote access.
- Examples: Internet, office LAN, college Wi-Fi, Bluetooth sharing.

- Uses of Computer Networks :

1.Resource Sharing – Share printers, files, software, and hardware among connected devices.

- One printer shared by all computers in an office.

2.Communication – Send emails, chats, video conferencing, and social networking.

- Sending emails or chatting through WhatsApp/Facebook.

3.Data Sharing – Store and access data from shared databases or cloud services.

- Google Drive or Dropbox used to share files.

4.Remote Access – Work from anywhere by connecting to systems remotely.

- Employees working from home via VPN/Remote Desktop.

5.Entertainment – Online games, movies, music, and streaming platforms.

- Watching movies on Netflix or playing PUBG online.

6.E-commerce & Online Services – Online shopping, banking, reservations, etc.

- Shopping on Amazon or doing online banking with Paytm.

Types of Computer Networks

1.Personal Area Network (PAN):

- A Personal Area Network (PAN) is the smallest type of computer network.
- It is designed for connecting devices within a short range (usually 1–10 meters).
- PAN is mainly used for personal purposes like sharing data, internet, or media between personal devices.

- Key Features of PAN:
  - Range: Very small (a few meters).
  - Devices: Smartphones, laptops, tablets, printers, smartwatches, headphones, etc.
  - Technology Used: Bluetooth, Infrared (IR), USB, Personal Hotspot.
  - Purpose: Quick communication and resource sharing between personal devices.
- Examples of PAN:
  - Connecting a smartphone to Bluetooth earphones.
  - Sharing files between a laptop and mobile via Bluetooth.
  - Using a mobile hotspot to give internet to another device.
  - Syncing a fitness tracker (smartwatch) with a smartphone.

2.Local Area Network (LAN)

- A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area such as a home, office, school, or campus.
- LANs provide high-speed communication and allow sharing of resources like files, printers, and internet connections.

- Examples of LAN:
  - A college computer lab where all PCs are connected to a central server.
  - An office network where employees share printers and files.
  - Home Wi-Fi connecting mobile phones, laptops, and smart TVs.

- Key Features of LAN:
- Range: Small area (a room, building, or campus).
- Speed: Very high (up to 1 Gbps or more).
- Devices: Computers, laptops, printers, routers, switches.
- Technology Used: Ethernet cables, Wi-Fi.
- Purpose: To share data, resources, and internet efficiently within a small area.

3.Metropolitan Area Network (MAN):

- A Metropolitan Area Network (MAN) is a network that covers a larger area than LAN, typically a city or a large campus.
- MANs are designed to connect multiple LANs within a metropolitan area to share resources and provide high-speed connectivity.

- Examples of MAN:
  - City-wide Wi-Fi networks provided by municipal authorities.
  - Cable TV networks connecting multiple offices and homes in a city.
  - University networks linking multiple campuses in the same city.

- Key Features of MAN:
- Range: Covers a city or a large campus (usually 5–50 km).
- Speed: Moderate to high (depends on technology).
- Devices: Routers, switches, computers, servers.
- Technology Used: Fiber optic cables, high-speed wireless, microwave links.
- Purpose: To connect several LANs in a city for communication and resource sharing.

4.Wide Area Network (WAN):

- A Wide Area Network (WAN) is a network that covers a large geographical area, such as a country, continent, or even worldwide.
- WANs are used to connect multiple LANs and MANs over long distances, enabling global communication and resource sharing.

- Examples of WAN:
  - The Internet – connecting millions of computers worldwide.
  - Bank networks – connecting branches of a bank across cities or countries.
  - Multinational companies – connecting offices in different countries for communication and data sharing.

- Key Features of WAN:
- Range: Very large, spanning cities, countries, or continents.
- Speed: Moderate (depends on technology and infrastructure).
- Devices: Routers, switches, satellites, servers, computers.
- Technology Used: Fiber optic cables, satellite links, leased lines, internet.
- Purpose: Long-distance communication, connecting multiple LANs/MANs globally.

Internetworks

- An internetwork is a collection of two or more computer networks that are connected together so that devices in different networks can communicate.
- It allows sharing of resources, data, and communication across networks that may have different technologies.
- Key devices used: Routers, gateways, switches.
- Example:
  - Connecting an office LAN with a branch LAN via the Internet.
  - The Internet itself is the largest and most famous internetwork.

?ARPANET (Advanced Research Projects Agency Network)

- ARPANET was the first operational packet-switching network.
- Developed by the U.S. Department of Defense in 1969.
- It was created to connect research institutions and allow resource sharing and communication between computers over long distances.
- It laid the foundation for the modern Internet.

- Key Features:
  - Used packet-switching technology instead of traditional circuit-switching.
  - Initially connected 4 computers at UCLA, Stanford, UCSB, and University of Utah.
  - Allowed remote access to computers and shared data between research centers.
- Example:
  - A researcher at UCLA could send a message or access a computer at Stanford University over ARPANET.
  - This was the beginning of global network communication that eventually evolved into the Internet.

? NSFNET (National Science Foundation Network)

- Introduction:
  - NSFNET was a program launched by the National Science Foundation (NSF), USA in 1985.
  - Its goal was to connect supercomputing centers across the United States and provide a high-speed backbone network for research and education.
  - It acted as a major milestone in the development of the Internet after ARPANET.

- Objectives of NSFNET
- To connect research and education institutes with high-speed networking.
- To provide access to supercomputers funded by NSF.
- To promote the development of a nationwide internet backbone.
- To encourage regional and campus networks to connect.
- To make advanced networking available to universities, laboratories, and researchers.

- Architecture of NSFNET:
  - NSFNET was designed as a three-tier architecture:
  - Backbone Network:
    - The high-speed core that linked supercomputer centers.
    - Initially operated at 56 kbps, later upgraded to 1.5 Mbps (T1) in 1988 and 45 Mbps (T3) in 1991.
  - Regional Networks:
    - Intermediate networks connecting local universities and colleges to the backbone.
    - Example: MIDnet, SURAnet, BARRnet.
  - Campus Networks:
    - Local networks within universities, colleges, or research institutes.
    - These connected faculty, students, and researchers to the regional network.

Internet Architecture

- Introduction:
  - The Internet architecture is the design framework that defines how different networks interconnect and how data flows between them.
  - It is based on the TCP/IP protocol suite, which ensures reliable communication.
  - The architecture is layered → each layer has a specific role and interacts only with the layers directly above and below it.

- Components of Internet Architecture
  - 1. End Systems (Hosts):
    - Devices at the edge of the Internet.
    - Examples: PCs, laptops, mobile phones, servers, IoT devices.
    - These run applications like email, web browsing, video streaming.
  - 2. Access Networks:
    - Connect end systems to the first router in the Internet.
    - Examples: Wi-Fi, DSL, Cable Modem, Fiber optics, Cellular 4G/5G, Ethernet LANs.
  - 3. Routers:
    - Special devices that forward packets between networks.
    - They form the backbone of the Internet.

- 4. Backbone Networks (ISPs):
  - High-capacity networks operated by Internet Service Providers (ISPs).
  - Organized in tiers:
    - Tier 1 ISPs → Large global providers (AT&T, NTT, Level 3).
    - Tier 2 ISPs → Regional providers.
    - Tier 3 ISPs → Local providers.

- 5. Protocol Stack (TCP/IP Model):
  - The Internet follows a layered architecture.
  - Internet Layer Model:

1. Application Layer
   - Provides services to users.
   - Examples: HTTP (web), SMTP (email), FTP, DNS.
2. Transport Layer:
   - Provides end-to-end communication between applications.
   - Examples: TCP (reliable), UDP (fast, connectionless).
3. Network Layer (IP Layer):
   - Responsible for routing and addressing.
   - Protocol: IP (IPv4, IPv6).

5.Data Link Layer:

- Transfers data between directly connected nodes.
- Examples: Ethernet, Wi-Fi, PPP.

6.Physical Layer:

- The actual hardware medium for transmission.
- Examples: Copper wire, Fiber optics, Radio waves.

- 6. DNS (Domain Name System):
  - Acts as the phonebook of the Internet.
  - Converts human-readable names (e.g., www.google.com) into IP addresses (e.g., 142.250.183.100).
- 7. Client-Server & P2P Architecture:
  - Client-Server → Clients (users) request services, servers respond (e.g., web browsing).
  - Peer-to-Peer (P2P) → Systems share resources directly (e.g., BitTorrent, file sharing).

Connection-Oriented & Connectionless Services

- 1. Connection-Oriented Service:
  - Works like a telephone call → connection must be established before data transfer.
    - Steps involved:
      - Connection Establishment
      - Data Transfer
      - Connection Release
  - Provides reliable delivery (no data loss, no duplication, proper sequencing).
  - Example: TCP (Transmission Control Protocol).
  - Use Cases: File transfer (FTP), Email (SMTP), Web browsing (HTTP over TCP).

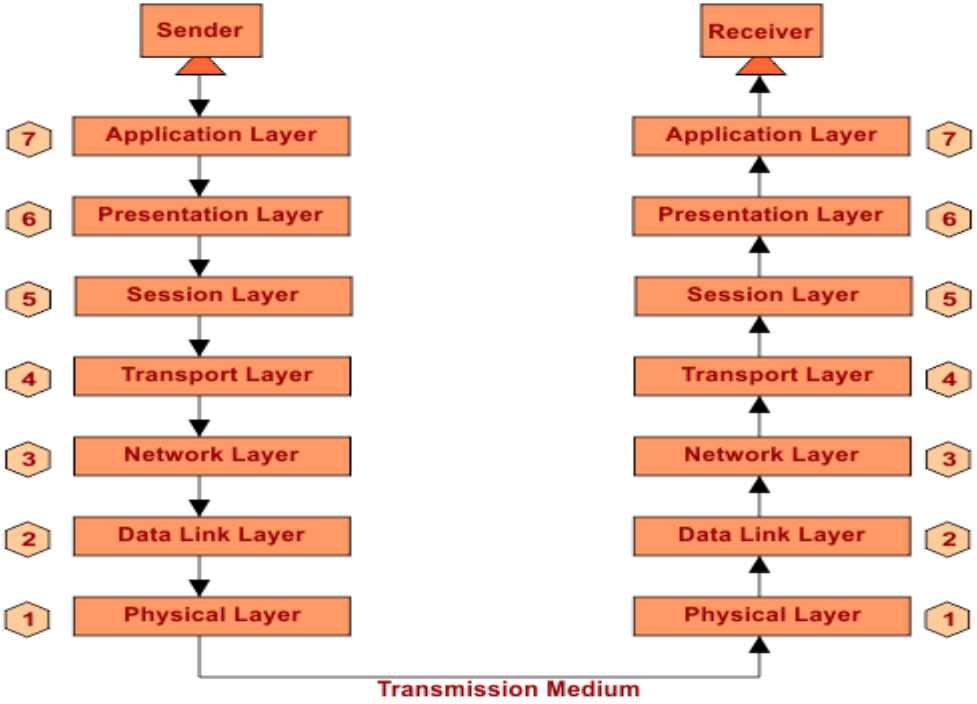
- 2. Connectionless Service:
  - Works like sending a letter/postcard → no prior connection setup.
  - Each packet is treated independently.
  - Less reliable (packets may be lost, duplicated, or arrive out of order).
  - Example: UDP (User Datagram Protocol).
  - Use Cases: Video streaming, VoIP, Online gaming (where speed is more important than reliability).

Service Primitives:

- Service primitives are the basic operations (commands) provided to the user/application to access network services.
- Think of them as the "buttons" an application presses to use the network.
- Common Service Primitives
  - LISTEN → Wait for an incoming connection (used by server).
  - CONNECT → Establish a connection to another user.
  - SEND → Transmit data.
  - RECEIVE → Accept incoming data.
  - DISCONNECT → Terminate the connection.
- Example:
  - A TCP server uses LISTEN → RECEIVE → SEND → DISCONNECT.
  - A TCP client uses CONNECT → SEND → RECEIVE → DISCONNECT.

- Relationship of Services to Protocols:
  - Service = What the network provides.
  - Protocol = How it is implemented.
  - Example:
  - Service: Reliable data transfer between two processes.
  - Protocol: TCP (defines rules, handshakes, acknowledgments, retransmissions).
  - Key Points:
  - Services are abstract → They define what is done.
  - Protocols are concrete → They define how it is done.
  - A service may be implemented using one or more protocols.

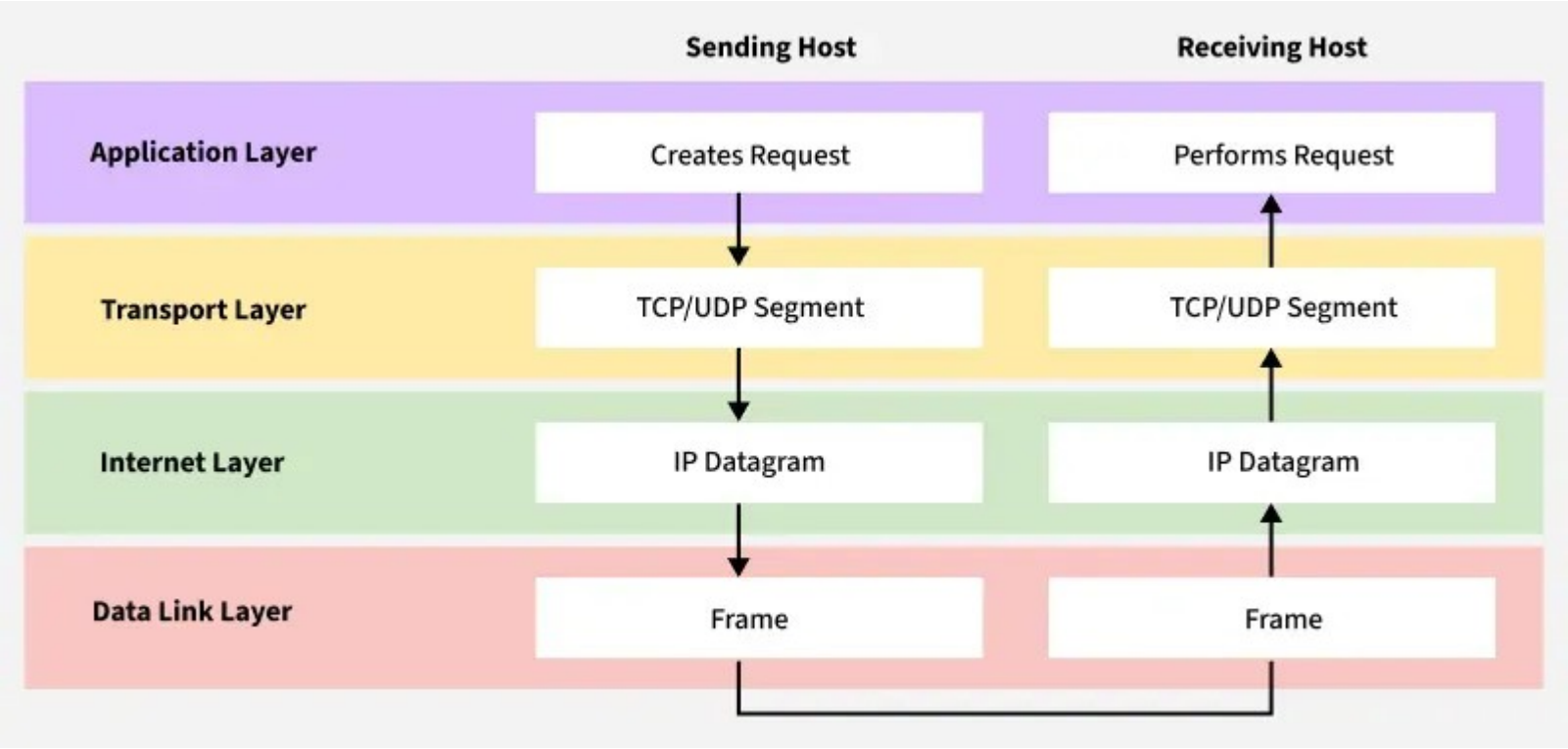
OSI Model


Sender
Receiver
7
Application Layer
Application Layer
7
6
Presentation Layer
Presentation Layer
6
5
Session Layer
Session Layer
5
4
Transport Layer
Transport Layer
4
3
Network Layer
Network Layer
3
2
Data Link Layer
Data Link Layer
2
1
Physical Layer
Physical Layer
1
Transmission Medium

OSI Model

07 Application Layer	→	The closest layer to the user; provides application services.
06 Presentation Layer	→	Encrypts, encodes and compresses usable data.
05 Session Layer	→	Establishes, manages, & terminates sessions between end nodes.
04 Transport Layer	→	Transmits data using transmission protocols including TCP & UDP.
03 Network Layer	→	Assigns global addresses to interfaces and determines the best routes through different networks.
02 Data link Layer	→	Assigns local addresses to interfaces, delivers information locally, MAC method
01 Physical Layer	→	Encodes signals, cabling and connectors, physical specifications.

TCP/IP Protocol



Layers of TCP/IP Model

- 1. Application Layer
  - The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network.
  - It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.
  - It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses).
  - It also manages things like data formatting, so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

- 2. Transport Layer
  - The Transport Layer is responsible for making sure that data is sent reliably and in the correct order between devices. It checks that the data you send like a message, file, or video arrives safely and completely.
  - This layer uses two main protocols: TCP and UDP, depending on whether the communication needs to be reliable or faster.
  - TCP is used when data must be correct and complete, like when loading a web page or downloading a file.
  - It checks for errors, resends missing pieces, and keeps everything in order. On the other hand, UDP (User Datagram Protocol) is faster but doesn’t guarantee delivery useful for things like live video or online games where speed matters more than perfect accuracy.

- 3. Internet Layer
  - The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device.
  - This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.
  - The main job of this layer is routing deciding the best way for data to travel.
  - It also takes care of packet forwarding (moving data from one point to another), fragmentation (breaking large data into smaller parts), and addressing.

- 4. Network Access Layer
  - The Network Access Layer is the bottom layer of the TCP/IP model. It deals with the actual physical connection between devices on the same local network like computers connected by cables or communicating through Wi-Fi.
  - This layer makes sure that data can travel over the hardware, such as wires, switches, or wireless signals.
  - It also handles important tasks like using MAC addresses to identify devices, creating frames (the format used to send data over the physical link), and checking for basic errors during transmission.

- When Sending Data (From Sender to Receiver):
  - Application Layer: Prepares user data using protocols like HTTP, FTP, or SMTP.
  - Transport Layer (TCP/UDP): Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
  - Internet Layer (IP): Adds IP addresses and decides the best route for each packet.
  - Link Layer (Network Access Layer): Converts packets into frames and sends them over the physical network.

- When Receiving Data (At the Destination):
  - Link Layer: Receives bits from the network and rebuilds frames to pass to the next layer.
  - Internet Layer: Checks the IP address, removes the IP header, and forwards data to the Transport Layer.
  - Transport Layer: Reassembles segments, checks for errors, and ensures data is complete.
  - Application Layer: Delivers the final data to the correct application (e.g., displays a web page in the browser).

Thank You